THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 2: **Polynomials** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $x = 5$ is a zero of $x^2 - 11x + k$, then k is: 1

(a) 30 **(b)** 31 **(c)** 29 **(d)** 32

2. The value of k for which $x = 5$ is a zero of $x^2 + kx - 15$ is: 1

(a) -3 **(b)** 0 **(c)** -1 **(d)** -2

3. The remainder when $3x^3 - 2x^2 + x - 6$ is divided by $x - 2$ is: 1

(a) -12 **(b)** 12 **(c)** -40 **(d)** 13

4. If the sum of the zeroes of $x^2 - kx + 21$ is 10, then the zeroes could be 3 and: 1

(a) 21 **(b)** 6 **(c)** 8 **(d)** 7

5. If α, β are zeroes of $x^2 - 2x - 15$, then $\alpha^2 + \beta^2$ is: 1

(a) 34 **(b)** 35 **(c)** 33 **(d)** 4

6. If α, β are zeroes of $x^2 - 9x + 20$, then $\frac{1}{\alpha} + \frac{1}{\beta}$ is: 1

(a) $\frac{20}{9}$ **(b)** $-\frac{9}{20}$ **(c)** $\frac{29}{20}$ **(d)** $\frac{9}{20}$

7. If $x = 6$ is a zero of $x^2 - 10x + k$, then k is: 1

(a) 23 **(b)** 25 **(c)** 26 **(d)** 24

8. The value of k for which $x = -3$ is a zero of $x^2 + kx - 12$ is: 1

(a) -1 **(b)** 1 **(c)** -2 **(d)** 0

9. The remainder when $x^3 + x^2 + x + 1$ is divided by $x - 2$ is: 1

(a) -15 **(b)** 15 **(c)** -5 **(d)** 16

10. If the sum of the zeroes of $x^2 - kx + 24$ is 11, then the zeroes could be 3 and: 1

(a) 24 **(b)** 9 **(c)** 8 **(d)** 7

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find a quadratic polynomial whose zeroes are 2α and 2β, where α, β are zeroes of $x^2 - 3x - 4$. 2

12. If α, β are zeroes of $x^2 - 10x + 9$, find $\alpha^2 + \beta^2$. 2

13. If $x = 5$ is a zero of $x^2 - 6x + k$, find k. 2

14. Find a quadratic polynomial whose zeroes are 2α and 2β, where α, β are zeroes of $x^2 - 8x + 15$. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. If α, β are zeroes of $x^2 - 11x + 18$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ and $\alpha^2 + \beta^2$. 3

16. Find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of $x^2 - 10x + 24$. 3

17. Divide $4x^4 + 2x^3 + x^2 - 3x + 7$ by $2x^2 + 1$ and write the quotient and remainder. 3

OR

If α, β are zeroes of $x^2 - 12x + 32$, find $\frac{1}{\alpha} + \frac{1}{\beta}$ and $\alpha^2 + \beta^2$. 3

18. Find a quadratic polynomial whose zeroes are the reciprocals of the zeroes of $x^2 - 9x + 8$. 3

OR

Divide $5x^4 - 4x^3 + 3x^2 + 2x - 1$ by $x^2 + x$ and write the quotient and remainder. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Find all the zeroes of $x^3 - 6x^2 + 5x + 12$ if two of its zeroes are 3 and 4. 5

OR

Find k if the product of the zeroes of $x^2 - 8x + k$ equals 3 times their sum, and find the polynomial. 5

20. Find all the zeroes of $x^3 - 4x^2 - 11x + 30$ if two of its zeroes are 2 and 5. 5

OR

Find k if the product of the zeroes of $x^2 - 9x + k$ equals 2 times their sum, and find the polynomial. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND